

Requirement Analysis

Introduction

Every successful software project begins with a clear understanding of its requirements. This phase focuses on identifying the problem, understanding user needs, and defining the features that the system must provide. For OptiCrop, the analysis revolves around solving one of agriculture's biggest challenges—selecting the right crop based on soil and environmental conditions.

Problem Analysis

Farmers often depend on traditional knowledge or assumptions while choosing crops. Factors such as soil nutrients, rainfall, temperature, humidity, and pH directly influence crop growth, but these parameters are rarely analyzed together before cultivation.

As a result, farmers may experience:

- Poor crop yield
- Loss of investment
- Soil nutrient imbalance
- Excessive fertilizer usage
- Inefficient water utilization

OptiCrop addresses these issues by converting agricultural data into intelligent crop recommendations using Machine Learning.

Requirement Identification

To solve the identified problem, the system should be capable of collecting agricultural data, processing it through a trained Machine Learning model, and generating accurate crop recommendations.

The application should also provide a simple interface that allows users to interact with the system without requiring technical knowledge.

User Requirements

The user should be able to:

- Register and log in securely.
 - Enter soil and environmental parameters.
 - Receive an instant crop recommendation.
 - View previous prediction records.
 - Submit feedback to improve future recommendations.
-

System Requirements

The system should be able to:

- Validate all user inputs.
 - Process agricultural parameters.
 - Execute the Machine Learning prediction model.
 - Display crop recommendations in a readable format.
 - Store prediction history.
 - Maintain user information securely.
-

Input Parameters

The prediction model requires the following agricultural values:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)

- Temperature
- Humidity
- Soil pH
- Rainfall

These parameters are sufficient to analyze soil fertility and environmental suitability for crop cultivation.

Expected Output

After processing the given inputs, the system should display:

- Recommended Crop
 - Prediction Status
 - Summary of Input Values
 - Stored Prediction Record
-

Functional Requirements

The main functions of OptiCrop include:

- User Authentication
- Crop Recommendation
- Prediction History
- Feedback Collection
- Database Management

Each module works together to provide a complete crop recommendation solution.

Non-Functional Requirements

The application should satisfy the following quality requirements:

Performance

Generate recommendations within a few seconds.

Accuracy

Produce reliable predictions using a trained Machine Learning model.

Security

Protect user accounts and stored information.

Usability

Provide a clean, responsive, and beginner-friendly interface.

Scalability

Allow future integration of weather forecasting, fertilizer recommendations, and disease detection.

Reliability

Operate continuously with minimal errors.

Technical Requirements

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Python

- Flask

Database

- MongoDB

Machine Learning

- Scikit-learn
- Pandas
- NumPy

Development Tools

- Visual Studio Code
 - Git
 - GitHub
-

Assumptions

The system assumes that users provide correct soil values and that the trained dataset accurately represents real agricultural conditions.

Limitations

- Prediction quality depends on the training dataset.
 - Incorrect input values may reduce accuracy.
 - Weather changes beyond historical data cannot always be predicted.
-

Summary

The Requirement Analysis phase transforms agricultural challenges into software requirements. By clearly defining user needs, system capabilities, and technical requirements, this phase provides the foundation for building an intelligent, reliable, and scalable crop recommendation platform.